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Comment

The problem with solutions: A commentary of "Blind alleys and fruitful pathways in the comparative study of cultural cognition" by Andrew Whiten

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Studying interspecific differences in social learning capacities for acquiring and cumulatively innovating solutions has, as Whiten (this issue) [1] points out, resulted in shifting goalposts for defining which capacities are uniquely human and which ones are not. A misguided (to our minds) yet widespread understanding, instantiated most explicitly by the *zone of latent solutions* view, is that CCE is best thought of in terms of a cumulative, ratchet-like form of exploration in a space of possible *solutions*. Artifacts and techniques (solutions) are modified and combined, increase in complexity and diversity, etc. It is recognized, for instance, that the space of possible solutions can expand through the harvesting of new phenomena [2]. The range of problems they solve, however, are generally ignored by the literature. Here we argue that many of the issues Whiten raises about CCE studies reside in this solution-based perspective, one that ignores the other side of the cumulative evolution coin: discovering, representing, and learning new problems.

The focus on solutions makes strong assumptions about the problems these solutions solve. In this view, the problem-space facing a species is fixed and ultimately does not require any special explanations: solutions evolve, the problems they solve don't. We disagree with this view. The extracts that Whiten presents from Teleki's [3] first-hand account of learning to fish termites illustrate why a myopic focus on solutions is unhelpful—a human researcher who has knowledge of the solution (having observed chimpanzees practice the technique) nonetheless struggles to acquire the skill until he gains sufficiently embodied experience of the problem parameters involved, such as the stringent constraints on tool pliability or the subtle motor manipulations required. In effect, his knowledge of the solution does not become usable until its corresponding problem can be represented.

For the human lineage, the problems solved through the CCE process are often not latent to the species: such problems build on and emerge from previous solutions within the species cultural repertoire but remain absent without them [4]. When Neanderthals first had the idea of combining a point with a shaft to produce a spear, the projected solution created a range of new problems to solve, such as finding how best to fit the haft so that the point would stick to the handle. Solving this problem led to new problems, one of which was figuring out how to balance the haft

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materials so that the spear would withhold the forces applied to it when thrown (or thrusted) onto (or into) a moving target. The haft problem is not something that existed prior to the invention of the composite tool, and the problem of having a resilient haft was not a problem that preexisted the solving of the haft problem.

Several cases of NHA traditions discussed by Whiten seem to be cumulative only in terms of the solutions they offer. Sasaki & Biro clearly demonstrate a case of incremental improvement in the solution to a navigational problem. However, in this case, as in others, the problem solved by the cumulative process is fixed from the start: how to minimize the distance traveled between two points. In such scenarios, while the solutions cumulatively improve, there is no change in the problem the species is solving. While we are happy to agree with Whiten that such a case is one of CCE, the pigeon case fails to satisfy the cumulative evolution of problems that is characteristic of human lineages. Rather than rejecting such cases as non-cumulative through shifting goalposts, we prefer to think of such cases as limited forms of CCE, namely, cumulative optimization. Cumulative optimization refers to a search process over the space of solutions and consists of generating and filtering for effective solutions at solving problems. Pigeons are capable of converging on an optimal flight path through cumulative transmission processes, but do not take insights about this optimization to generalize from the discovered solutions to solve novel problems [5].

In contrast, human CCE, at the species level, is open-ended, i.e., it is capable of taking culturally emerging problems and solving those together with the problems that those solutions generate. This open-ended growth of diversity and complexity requires optimization, but also another causally interacting search process: discovery. Discovery inverts the causality of the problem-solution process and constitutes a search process of exploring and exploiting novel problems, and then looking for solutions to those discovered problems. Open-ended CCE, such as that seen in humans, emerges from the interaction of optimization and discovery as individuals not only generate novel solutions to established problems, but also recognize and adapt to novel problems arising as a result of these solutions. Importantly, problems can exist as a culturally inherited informational content independent of their corresponding technical solutions (behaviors), and this independence is key to maintaining the recursive loop that drives cumulative effects. In other words, new problems emerge out of the cultural process, to which the cultural process then finds solutions, which in turn generates more problems, and so on and so forth.

Adopting a problem-based approach to studying cultural learning raises important methodological considerations, particularly for the studies with captive chimps that Whiten critiques in his paper, which aim to show that technical solutions observed in wild chimps are within the ZLS of individual chimpanzees to innovate. In these studies, experimentalists pay close attention to translating the solutions that individuals use from the wild to the lab. However, as Whiten frequently points out, in translating these solutions the researchers also simplify and make more salient the problems that individuals face: e.g., instead of fishing for mobile, even aggressive, termites from a hidden mound, chimpanzees are ostensively presented with an apparatus and a tool to fish for ketchup. Such experiments therefore show that chimpanzees can perform these action sequences in the lab as in the wild, but do not explain how chimpanzees came to recognize that such action sequences were relevant to the complex ecological problems that they solve.

Designing experiments that translate rich ecological problems into controllable lab problems is a difficult undertaking. Tackling it will require looking to qualitative approaches that prioritize a first-person understanding of the techniques being studied and the problems involved—Teleki's account offers an insightful starting point for comparative research, the field of experimental archaeology can offer rich explanations of the mechanical and practical constraints that delineate the problem space associated with many human techniques, and anthropological accounts of experts' own experiences can show how individuals actually conceive and construct this problem space. Understanding the problems that individuals face, and how they represent these problems to generate or learn solutions, is an important step in understanding both learning and invention of techniques.

The current emphasis on solutions paints an impressive picture of CCE abilities in humans, with plenty of depictions of tool forms gradually and cumulatively changing in shape over time (e.g., stone points, ship rudders). In contrast, non-humans seem to have a hard time coming up with increasingly complex solutions, and those they can produce seems in the range of individuals learning on their own. Whiten makes a good case that searching for interspecific differences in the capacity for cumulative cultural learning may not center on what we term solutions. Where we believe that he could go further is in considering how problems themselves are learned, inherited, or even invented.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Whiten Andrew. Blind alleys and fruitful pathways in the comparative study of cultural cognition. Phys Life Rev 2022;43:211–38. https://doi.org/10.1016/j.plrev.2022.10.003.
[2] Derex Maxime. Human cumulative culture and the exploitation of natural phenomena. Philos Trans R Soc B 2022;377(1843). https://doi.org/10.1098/rstb.2020.0311.
[3] Teleki Géza. Chimpanzee subsistence technology: materials and skills. J Hum Evol 1974;3:575–94. https://doi.org/10.1016/0047-2484(74)90018-9.
[4] Charbonneau Mathieu. All innovations are equal, but some more than others: (re)integrating modification processes to the origins of cumulative culture. Biol Theory 2015;10(4):322–35. https://doi.org/10.1007/s13752-015-0227-x.
[5] Winters James. Escaping optimization traps: the role of cultural adaptation and cultural exaptation in facilitating open-ended cumulative dynamics. Palgrave Commun 2019;5(149):1–13. https://doi.org/10.1057/s41599-019-0361-3.